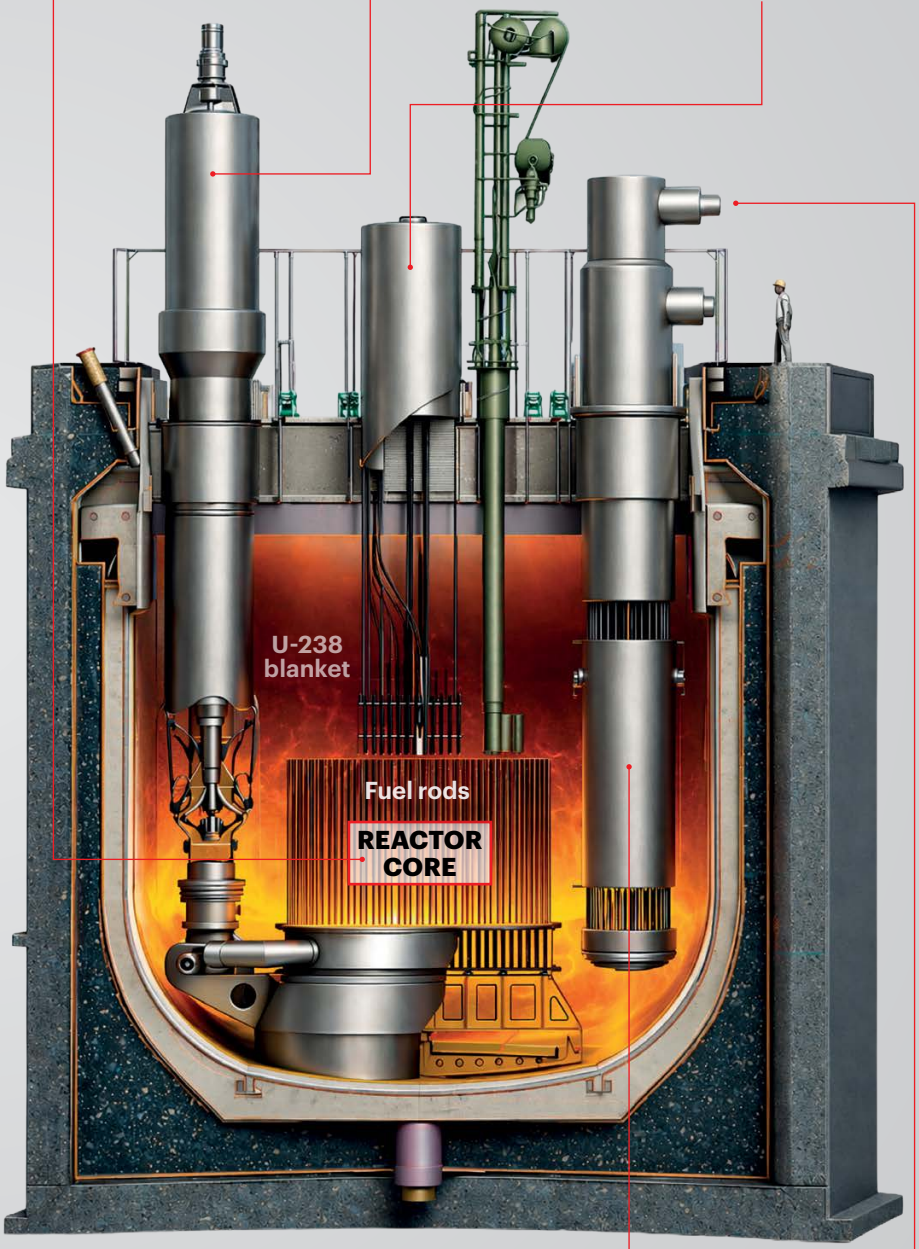
What goes on inside the reactor

Beyond the nuclear reaction, a fast breeder reactor relies on an intricate system that converts the heat it generates into electricity

1. Fast neutrons split Pu-239 to release heat while also breeding more Pu-239 from U-238. This allows the reactor to generate more fuel than it consumes

2. The reactor vessel is filled with liquid sodium coolant—used instead of water because it does not slow fast neutrons and transfers heat efficiently

3. Control rods absorb neutrons to regulate and stabilise the chain reaction



4. The heat exchanger transfers heat to a separate non-radioactive sodium loop. It is kept isolated since sodium reacts violently with air and water

5. The secondary sodium then produces steam, which spins a turbine to generate electricity

Sources: Bhabha Atomic Research Centre, Press Information Bureau (GoI) & International Panel on Fissile Materials

The Global Race

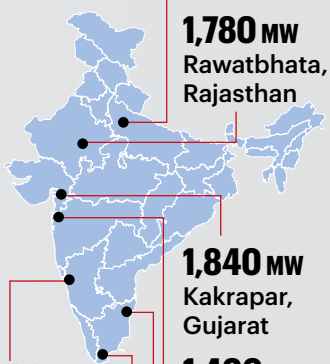
- Several countries pursued fast breeder reactors (FBRs) to extend uranium supplies and better manage spent fuel
- Interest waned as uranium proved more abundant; high costs and safety concerns limited adoption
- Japan's Monju reactor closed after a sodium leak in 1995; France shut down Superphénix, the first commercial FBR, in 1998
- Currently, only Russia operates commercial FBRs; China is also developing them

India's Nuclear Capacity

8,780 MW
Total installed capacity

440 MW
Narora, Uttar Pradesh

1,780 MW
Rawatbhata, Rajasthan



1,840 MW
Kakrapar, Gujarat

880 MW
Kaiga, Karnataka

1,400 MW
Tarapur, Maharashtra

440 MW
Kalpakkam, Tamil Nadu

2,000 MW
Kundakulam, Tamil Nadu

- India operates the world's sixth-largest, PHWR-dominant nuclear reactor fleet
- In FY25, they generated 56,681 million units—3 per cent of India's electricity
- The Nuclear Energy Mission aims at 100 GW of nuclear power capacity by 2047
- This will be achieved via 700 MWe indigenous PHWRs, larger imported reactors and small modular reactors